

VELTRIX AI

Multilingual Phishing & Threat Intelligence Platform

Hackathon Project Report • Team Vortex

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Problem Statement

Phishing attacks are growing in speed and complexity, yet most users lack real-time protection. Existing tools are fragmented, unexplainable, and inconsistent across platforms. Veltrix AI solves this with unified, AI-powered threat detection across email, web, and mobile.

SDG Alignment

SDG 9 — Industry, Innovation and Infrastructure (*Primary*)

SDG 16 — Peace, Justice and Strong Institutions (*Secondary*)

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In fulfillment of the hackathon project requirements, Team Vortex — Veltrix AI — v1.0.0



Abstract

Veltrix AI is a multi-platform phishing detection system designed to reduce user exposure to malicious emails, deceptive links, and social engineering attacks. The project combines machine learning, rule-based heuristics, and real-time client-side enforcement to deliver both predictive detection and immediate protective actions.

The system integrates five components: a FastAPI backend for analysis and orchestration, a Chrome extension for Gmail scanning, a Next.js dashboard for security visibility, a Flutter mobile app for manual scanning, and offline training pipelines for model management.

The core contribution is practical security workflow integration — detection is not presented as a standalone score, but as an end-user protection loop including explainable reasons, threat labels, URL-level risk context, and blocking actions directly in the user interface.

Problem Statement & Motivation

Phishing remains one of the most frequent and damaging cyber attack vectors because it exploits both technical weaknesses and human behavior. Attackers continuously adapt tone, urgency, and visual formatting to bypass intuition and standard filters.

Key Challenges

- Detection latency — most tools alert users after interaction, which is often too late.
- Fragmented tooling — email filtering, URL scanning, and dashboards exist as separate products.
- Low explainability — binary safe/unsafe outputs offer no reasoning for the user.
- Cross-platform inconsistency — users switch between desktop and mobile with no unified protection.

Veltrix AI addresses all four challenges by combining analysis, threat signaling, and user control into one coherent architecture that spans browser, dashboard, and mobile environments.

SDG Alignment

Veltrix AI maps to two United Nations Sustainable Development Goals that reflect its dual impact on technological innovation and digital safety.

Primary — SDG 9: Industry, Innovation and Infrastructure

Veltrix AI contributes to SDG 9 by building intelligent, resilient digital infrastructure through AI and machine learning. The platform demonstrates how innovative technology can be applied to solve real-world security challenges — making digital systems more robust, accessible, and trustworthy for everyday users across browser and mobile environments.



Secondary — SDG 16: Peace, Justice and Strong Institutions

Veltrix AI supports SDG 16 by actively combating cybercrime and digital fraud. Phishing attacks disproportionately target vulnerable individuals and undermine trust in digital institutions. By providing real-time threat detection, explainable warnings, and user-controlled blocking, Veltrix AI empowers users to engage safely online — contributing to a more just and secure digital society.

Methodology

1. Data Engineering & Model Training

The training pipeline sources labeled examples from multiple phishing, spam, smishing, and URL datasets. Processing includes schema-aware column mapping, text concatenation, label normalization, balanced sampling, and weighted dataset contributions.

TF-IDF with unigram and bigram features feeds into a Logistic Regression classifier with class balancing. Evaluation includes classification reports, confusion matrix analysis, ROC-AUC scores, and sanity test suites on real phishing phrases.

2. Backend Service Design

The FastAPI backend is organized into a routing layer, schema layer, configuration/store layer, and ML inference layer. It returns structured outputs including label, confidence score, threat reasons, URL risk objects, and blacklist flags — keeping all clients consistent.

3. Hybrid Detection Strategy

Veltrix AI combines ML classification for semantic phishing signals, URL risk heuristics for domain anomalies, blacklist overrides for user-defined policy, and fallback rule logic for resilience. Blacklisted entities can override neutral predictions to minimize false negatives.

4. Chrome Extension

A Manifest V3 Chrome extension serves as the primary in-browser protection layer. It scans Gmail content in real time, invokes backend analysis APIs, renders color-coded threat badges, and coordinates link blocking before user interaction.

5. Dashboard & Mobile App

The Next.js dashboard provides visibility into threat activity, blocked entities, and manual scanning workflows. The Flutter mobile app extends access to phishing checks in mobile contexts, including share-to-app from SMS and messaging platforms.



System Architecture

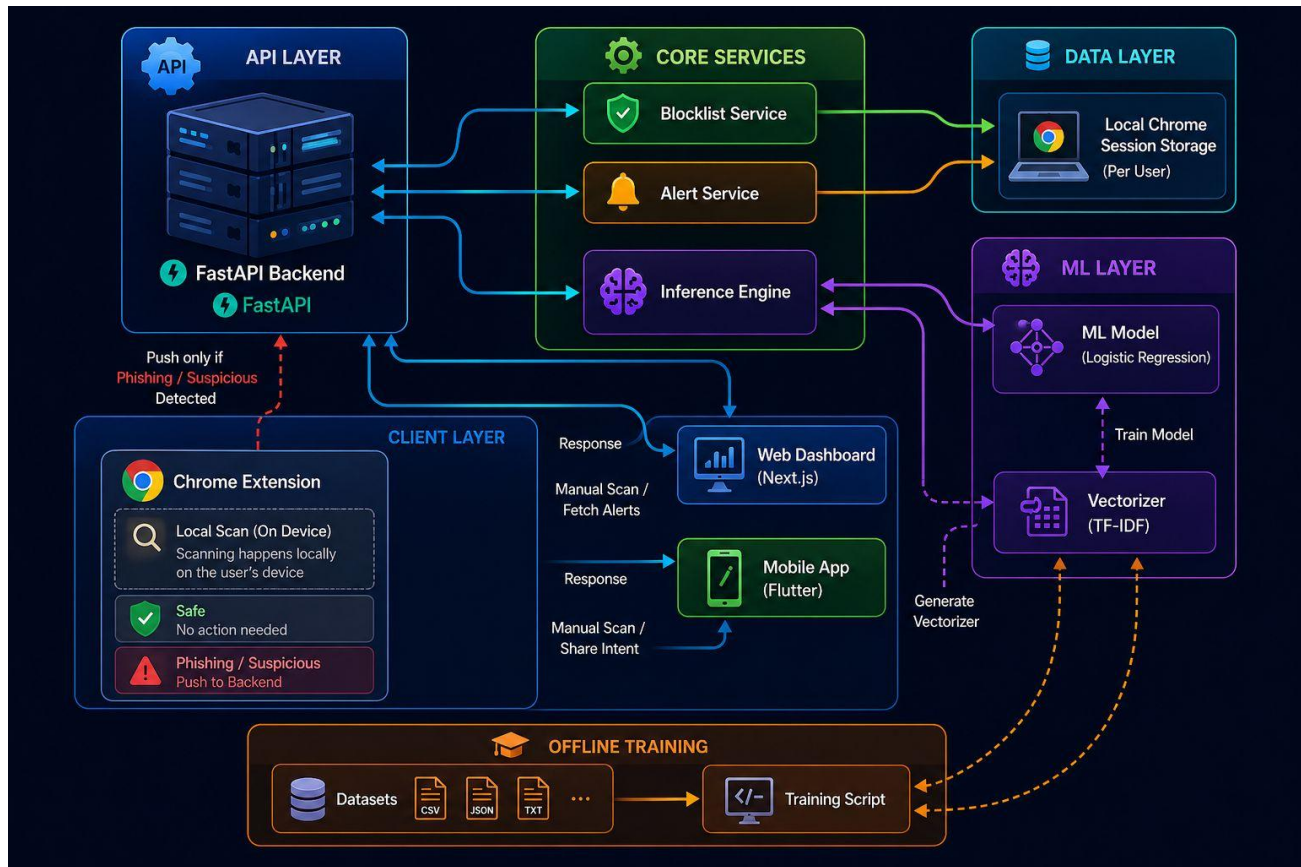


Figure 1: Veltrix AI System Architecture

Technology Stack

Layer	Technology
Backend API	FastAPI, Uvicorn, Pydantic, python-dotenv
ML / Detection	scikit-learn, TF-IDF, Logistic Regression
URL Analysis	tldextract, custom heuristics
Browser Extension	Chrome MV3, JavaScript, HTML, CSS
Dashboard	Next.js 14, React 18, TypeScript, Tailwind CSS
Mobile App	Flutter, Dart
Training Data	Multi-dataset phishing/spam corpus (CSV)



Key API Endpoints

Method	Endpoint	Description
GET	/health	Service and model status
POST	/analyze-text	Analyze email/body text
POST	/analyze-batch	Batch analysis
POST	/analyze-url	URL risk analysis
POST	/block-sender	Add sender to blocklist
GET	/alerts	Get recent threat alerts
GET	/blocked	Get blocked senders/URLs

Results

Functional Outcomes

- Working FastAPI backend with full analysis and policy endpoints.
- Persisted ML artifacts (model.pkl, vectorizer.pkl) for low-latency inference.
- Gmail Chrome extension with real-time detection and color-coded threat badges.
- Next.js web dashboard for security monitoring and review.
- Flutter mobile client with scan, alerts, and history screens.

Detection & Evaluation

- Classification report on held-out test data with precision, recall, and F1 metrics.
- Confusion matrix inspection to minimize false negatives in user-facing warnings.
- ROC-AUC computation for probability-based model quality review.
- Sanity test suite on common phishing phrases to validate model behavior.

Risk Scoring Model

The system generates a unified risk score (0–100) combining ML prediction, URL risk, keyword analysis, and sender reputation — mapped to three actionable tiers:

- 0–24 → Safe (green)
- 25–49 → Suspicious (amber)
- 50+ → Phishing (red) — immediate blocking

Security Workflow Outcomes

- End-to-end threat handling: from message inspection to user warning within the Gmail interface.
- Explainable outputs via threat reasons and confidence descriptors.
- User policy enforcement through persistent sender and URL blocklists.
- Alert generation for visibility and audit-level review in the dashboard.



Conclusion

Veltrix AI demonstrates a practical, integrated approach to phishing defense by unifying machine learning detection, heuristic URL risk analysis, real-time user warnings, and policy-driven blocklist controls. The project moves beyond isolated classification to deliver a complete security loop: analysis, explanation, enforcement, and visibility.

The implementation confirms that phishing defense is most effective when intelligence is coupled with immediate interface-level feedback across multiple platforms. By combining a robust backend with browser and mobile clients, Veltrix AI improves usability while maintaining a scalable technical foundation.

Key Insights

- Hybrid models outperform single-method detection for real-world phishing behavior.
- Explainability and in-context alerts are critical for user trust and safer decision-making.
- Modular architecture enables future growth: multilingual NLP, richer URL intelligence, and enterprise deployments.

Veltrix AI provides a strong foundation for continued development into a production-grade threat intelligence and anti-phishing platform.



Built at Nexora Hackathon 2026 • Team Vortex • v1.0.0